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Date of Performance-	16/03/2026

Practical No.: 7

Title: TELNET

Objective: To implement TELNET in a Network

Theory and Methodology: -

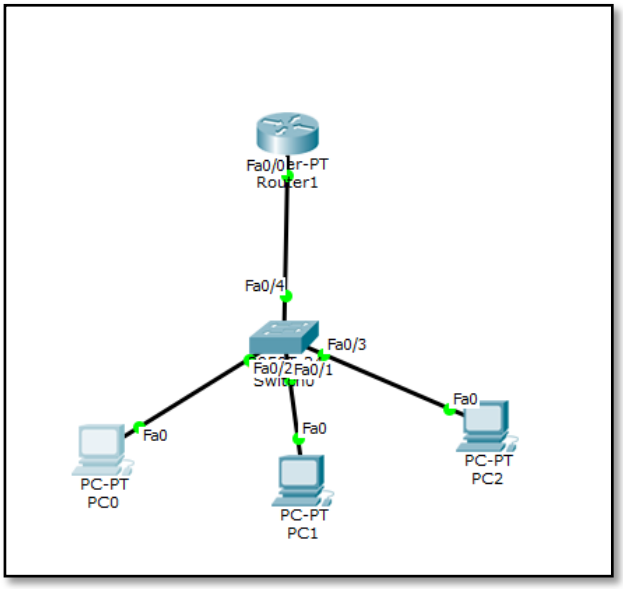
Telnet (Telecommunication Network) is a network protocol used to remotely access and manage devices over a TCP/IP network. It allows a user to log in to a remote computer, router, or switch and execute commands as if working directly on that device.

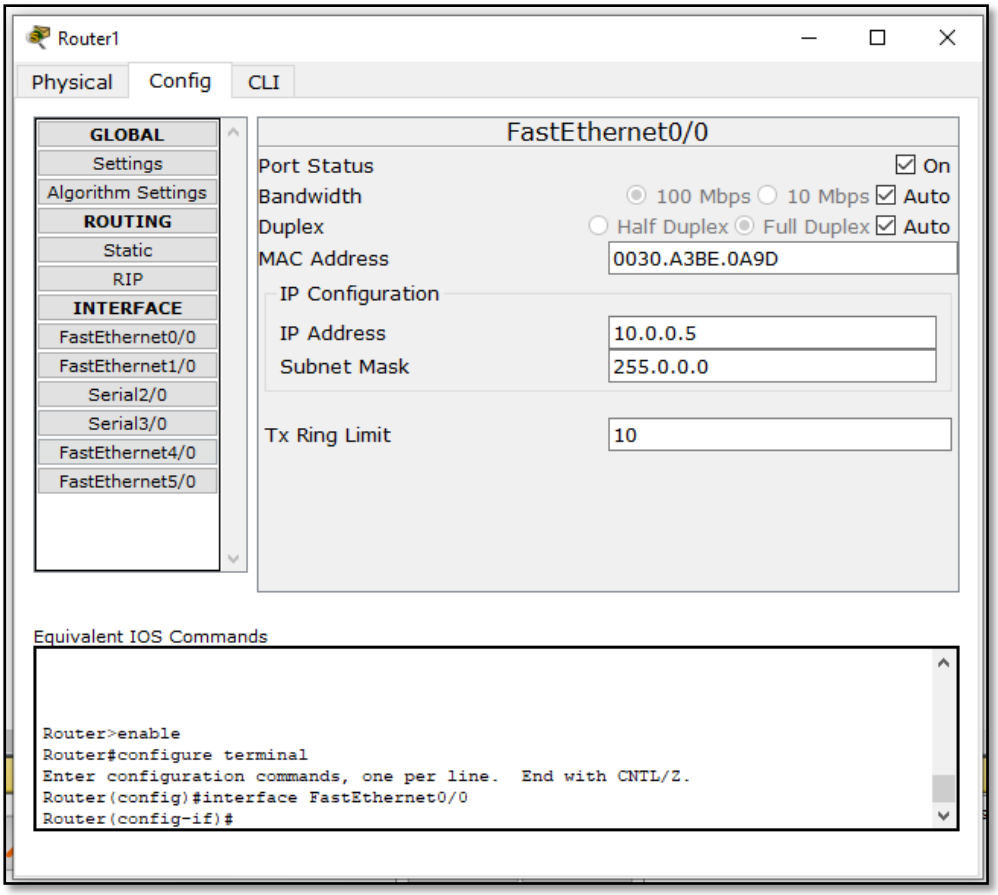
Telnet operates on TCP port 23 and provides a command-line interface (CLI) for communication between the client and server. It follows a client-server architecture, where the Telnet client sends requests and the Telnet server processes them and returns responses.

In computer networks, Telnet is commonly used for:

- Remote configuration of routers and switches
- Network troubleshooting
- Monitoring network devices
- Managing servers from a remote location

Results: - Screenshots-






```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ex
Router(config)#interface VLAN 1
                          ^
% Invalid input detected at '^' marker.

Router(config)#ex
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface VLAN 1
                          ^
% Invalid input detected at '^' marker.

Router(config)#interface VLAN 2
                          ^
% Invalid input detected at '^' marker.

Router(config)#
Router(config)#interface FastEthernet0/0
Router(config-if)#exs
                 ^
% Invalid input detected at '^' marker.

Router(config-if)#ex
Router(config)#enable password admin2
Router(config)#line vty 0 15
Router(config-line)#password CISCO2
Router(config-line)#login
Router(config-line)#
```

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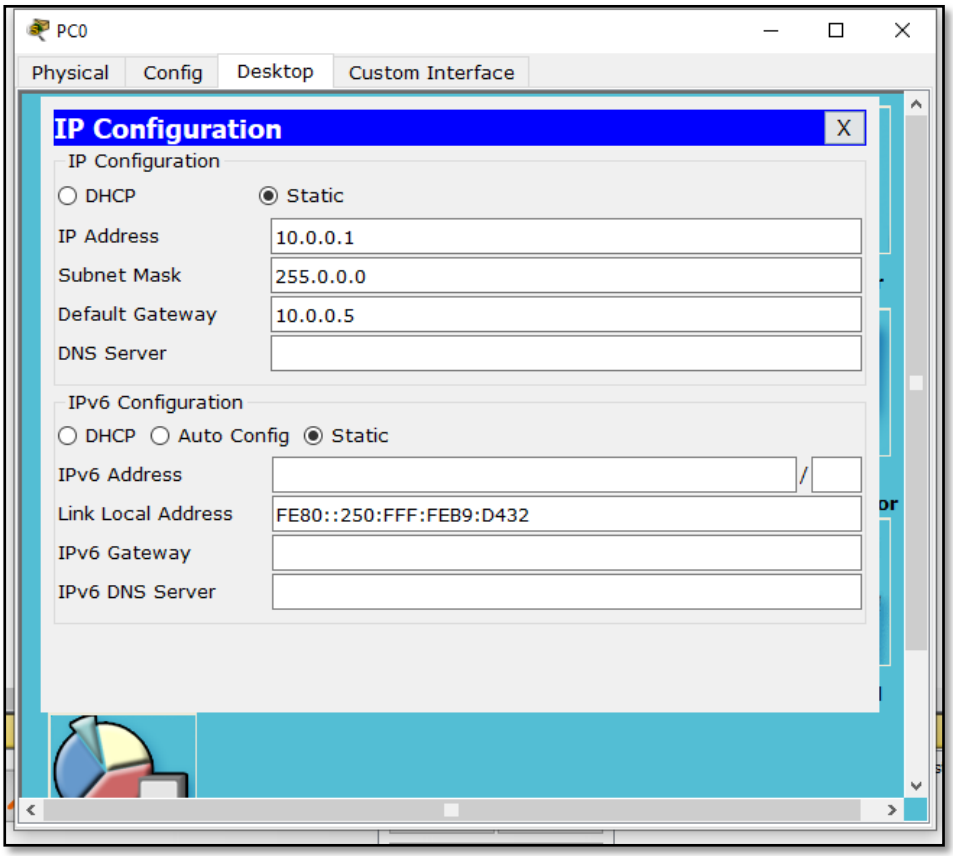
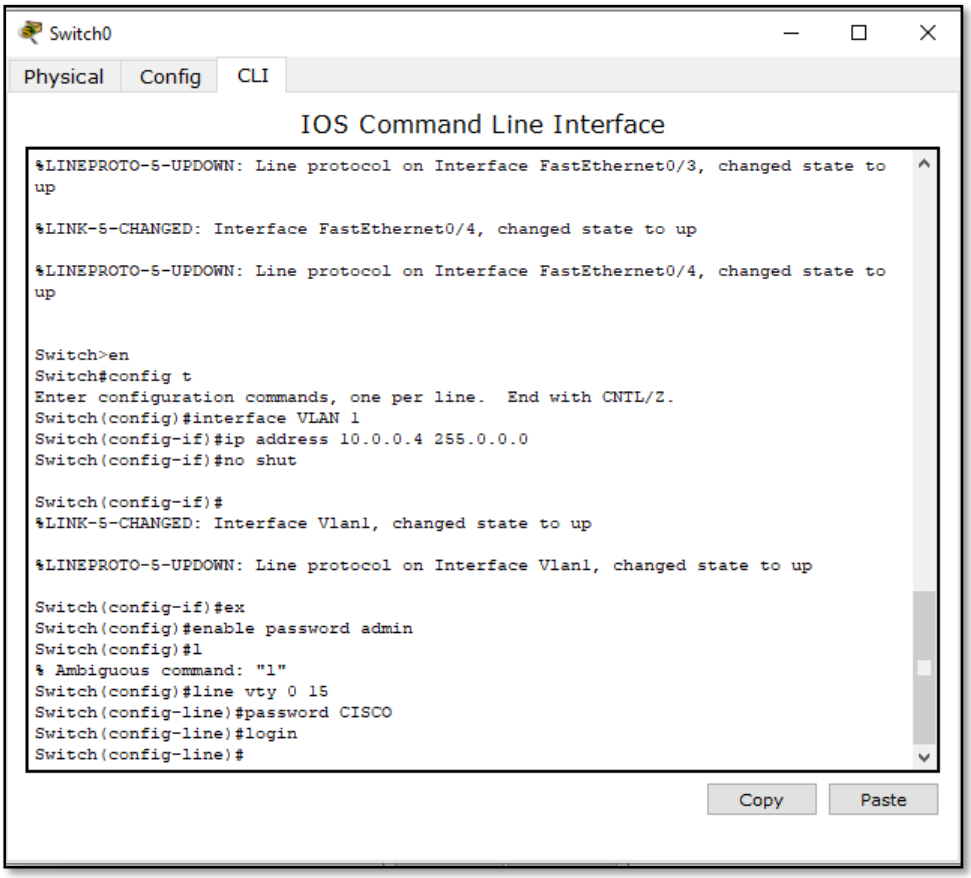

PC0
Physical
Config
Desktop
Custom Interface
IP Configuration
X
IP Configuration
DHCP
Static
IP Address
10.0.0.1
Subnet Mask
255.0.0.0
Default Gateway
10.0.0.5
DNS Server
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
Link Local Address
FE80::250:FFF:FEB9:D432
IPv6 Gateway
IPv6 DNS Server


Switch0
Physical
Config
CLI
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface VLAN 1
Switch(config-if)#ip address 10.0.0.4 255.0.0.0
Switch(config-if)#no shut
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
Switch(config-if)#ex
Switch(config)#enable password admin
Switch(config)#l
% Ambiguous command: "l"
Switch(config)#line vty 0 15
Switch(config-line)#password CISCO
Switch(config-line)#login
Switch(config-line)#
Copy
Paste

Command Prompt

```
Packet Tracer PC Command Line 1.0
PC>telnet 10.0.0.4
Trying 10.0.0.4 ...Open


User Access Verification

Password:
Switch>en
Password:
Switch#show MAC-ADDRESS-TABLE
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0030.a3be.0a9d    DYNAMIC     Fa0/4
   1    0050.0fb9.d432    DYNAMIC     Fa0/2
Switch#
```

Command Prompt

```
Packet Tracer PC Command Line 1.0
PC>telnet 10.0.0.5
Trying 10.0.0.5 ...Open


User Access Verification

Password:
Password:
Router>en
Password:
Router#show MAC-ADDRESS-TABLE
            ^
% Invalid input detected at '^' marker.

Router#show MAC-ADDRESS-TABLE
            ^
% Invalid input detected at '^' marker.

Router#show iproute
           ^
% Invalid input detected at '^' marker.

Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, FastEthernet0/0
Router#
```

Conclusion: -

In this experiment, TELNET was successfully implemented using Cisco Packet Tracer to establish remote communication between network devices. The practical demonstrated how TELNET works on TCP port 23 using a client-server architecture, allowing a user to access and configure a remote device through the command-line interface. It helped in understanding how network administrators can remotely manage routers and switches, perform troubleshooting, and monitor network performance. The experiment also highlighted the working procedure of enabling TELNET on a device and verifying connectivity across the network. Overall, this practical provided hands-on experience with remote device management and improved understanding of TELNET protocol functionality in computer networks.

Answer for the following Questions-

1) Telnet sessions drop after 5 minutes of inactivity.
Ans: This is typically caused by an **exec-timeout** setting on the VTY (virtual terminal) lines of the router or switch. For example, the command exec-timeout 5 0 terminates idle sessions after 5 minutes. It could also be caused by a TCP keepalive setting or an intermediate firewall timing out idle connections.

2) In a lab, students Telnet to 192.168.10.1 expecting a router, but reach a different device.
Ans: Possible causes:

- **Duplicate IP address** – Another device on the network is configured with the same IP (192.168.10.1).
- **Wrong device connected** – The cable from the student's PC is plugged into a different device (e.g., a switch or another router).
- **NAT or port forwarding** – The network translates 192.168.10.1 to a different internal device.
- **Proxy or terminal server** – An intermediate device intercepts Telnet connections.

3) What is Telnet used for?
Ans: Telnet is used for **remote command-line access** to network devices (routers, switches, servers) over a network. It allows a user to log in and execute commands as if physically connected to the device's console port.

4) Mention two security disadvantages of Telnet.
Ans:

- **No encryption** – All data, including usernames and passwords, is sent in clear text.
- **No authentication or integrity checks** – Vulnerable to man-in-the-middle attacks, session hijacking, and credential sniffing.

5) A user cannot Telnet to a router even though it responds to ping. List possible reasons.
Ans:

- **Telnet not enabled** – The router's VTY lines are not configured to accept Telnet (or only SSH is allowed).
- **Access list blocking Telnet** – An ACL applied to VTY lines denies the user's IP address.
- **Transport input mismatch** – The VTY lines have transport input ssh only, or transport input none.

- **No login or password configured** – VTY lines lack a password or AAA authentication.
- **Incorrect VTY line range** – All available VTY lines (0–4 or 0–15) are already in use.
- **Firewall or ACL on the path** – An intermediate device blocks TCP port 23 (Telnet) while allowing ICMP (ping).